SAMRAT ASHOK TECHNOLOGICAL INSTITUTE
(Engineering College),VIDISHA (M.P.)
(An Autonomous Institute Affiliated to RGPV Bhopal)

Department of Information Technology

Semester/Year			Program		B. Tech.				
Subject Category	DLC	Subject Code:	IT26102P	Subject Name:	Linux Lab				

Maximum Marks Allotted						Contact Hours			Total Credits
Theory			Practical		TotalMarks				
End Sem	Mid-Sem	Assignment/ Quiz	End Sem	Lab- Work		L	T	P	
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Pre-requisites:

Course Objective:

1. To teach principles of operating system including File handling utilities, Security by file permissions, Process utilities, Disk utilities, Networking Commands, Basic Linuxcommands, Scripts and filters.
2. To familiarize fundamentals of the Bourne again shell (bash), shell programming, pipes, input and output redirection Control structures, arithmetic in shell interrupt processing, functions, debugging shell scripts.
3. To impart fundamentals of file concepts kernel support for file, File structure related system calls (file API's).
4. To facilitate students in understanding Inter-process communication.
5. To facilitate students in understanding semaphore and shared memory.
6. To facilitate students in understanding process.

Course Outcomes:

Upon completion of this course, the student will be able to:

CO1. Ability to use various Linux commands that are used to manipulate system operations at admin level and a pre-requisite to pursue job as a Network administrator.

CO2. Ability to write Shell Programming using Linux commands.

CO3. Ability to design and write application to manipulate internal kernel level Linux FileSystem.

CO4.Ability to develop IPC-API's that can be used to control various processes for synchronization.

CO5. Ability to develop Network Programming that allows applications to make efficient use of resources available on different machines in a network.

UNITs	Descriptions	CO's
I	**INTRODUCTIONTOLINUXANDLINUXUTILITIES:**Abriefhistory of LINUX, architecture of LINUX, features of LINUX, introduction to vi editor. Linux commands, file handling utilities, security by file permissions, process utilities, disk utilities, networking commands, Text Processing utilities and backup utilities	CO1
II	**Introduction to Shells**: Linux Session, Standard Streams, Redirection, Pipes, Command-Line Editing, Options, Shell/Environment Customization. **Filters:** Filters and Pipes, Concatenating files, Display Beginning and End of files, Operations on Characters, Files with Duplicate Lines, Count Characters, Words or Lines, Comparing Files.	CO2
III	**Grep:** Operation, grep Family, Searching for File Content. **Sed**: Scripts, Operation, Addresses, commands, Applications, grep and sed. UNIX FILE STRUCTURE: Introduction to UNIX file system, inode (Index Node), file descriptors, system calls and device drivers. **File Management**: File Structures, System Calls for File Management, Directory API.	CO3
IV	**PROCESS AND SIGNALS**: Process, process identifiers, process structure: process table, viewing processes, system processes, process scheduling, zombie processes, orphan process, unreliable	

		CO4

	signals, interrupted system calls. **File locking**: creating lock files, locking regions, use of read and write with locking, competing locks, other lock commands, deadlocks.	
V	**INTER PROCESS COMMUNICATION**: Pipe, process pipes, thepipe call, parent and child processes, and named pipes, semaphores, message queues, shared memory. **INTRODUCTION TO SOCKETS**: Socket, socket connections-socket attributes, socket addresses.	CO5
Guest Lectures(if any)		
Total Hours		

Text Books-

1. W. Richard. Stevens (2005), Advanced Programming in the UNIX Environment, 3rd edition, Pearson Education, New Delhi, India.
2. UNIX and shell Programming Behrouz A. Forouzan, Richard F. Gilberg. Thomson

REFERENCES Books-:

1. Linux System Programming, Robert Love, O' Reilly, SPD.
2. Advanced Programming in the UNIX environment, 2nd Edition, W.R.Stevens, Pearson Education.
3. UNIX Network Programming, W.R. Stevens, PHI. UNIX for Programmers and Users,3rd Edition, Graham Glass, King Ables, Pearson Education

Modes of Evaluation and Rubric

The evaluation modes consist of performance in Quiz/ Assignments, term work, and end-semester Practical examinations.

List/Links of e-learning resource

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO
CO-1	3	2	1		3			1	1	1		2
CO-2	3	3	2	1	3				1	1		2
CO-3	3	3	3	2	3			1	1	1		2
CO-4	3	3	3	2	3			1	2	1		2
CO-5	3	3	3	2	3	1	1	1	2	2	1	2

List of Experiments

1. Write a program using echo, printf, script, passwd, uname, who, date, stty, pwd commands.
2. Write a program using unlink, du, df, mount, umount, find, unmask, ulimit, ps, w, finger, arp, ftp commands.
3. Write a program using telnet, rlogin, Text Processing utilities and backup utilities, tail, head ,sort,nl,uniq,grep,egrep,fgrep,cut,paste,join,tee,pg,comm,cmp,diff,tr,awk commands.
4. Write a shell script that accepts a file name, starting and ending line numbers as arguments and displays all the lines between the given line numbers.
5. Illustrate by writing script that will print, message "Hello World, in Bold and Blink effect, and in different colours like red, brown etc using echo commands?
6. Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.
7. Illustratebywritingscriptusingforlooptoprintthefollowingpatterns?
8. Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.
9. Write a program inter-process communication.
10. Write a program to communicate using sockets.

Recommendation by Board of studies on	
Approval by Academic council on	
Designed by	IT Department